

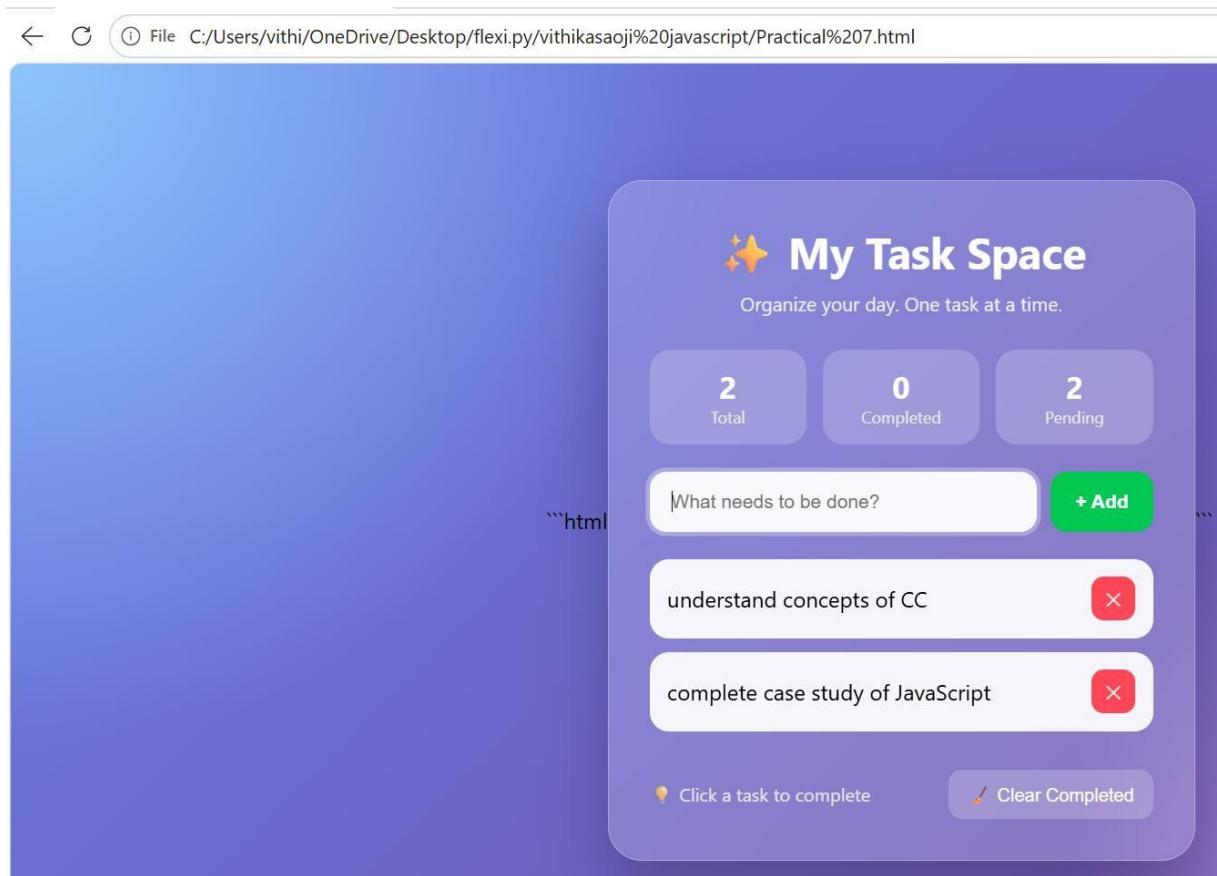
Practical No.: 7

1. Title: Perform DOM traversal and updates; develop a to do list app with add, edit, and delete features.

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: The Document Object Model (DOM) represents an HTML webpage as a tree of elements that can be accessed and modified using JavaScript. DOM traversal is the process of navigating through HTML elements using properties and methods such as `parentElement`, `children`, `nextElementSibling`, `querySelector()`, and `querySelectorAll()`. DOM updates allow JavaScript to dynamically change the content, style, and structure of a webpage without reloading it. In a To-Do List application, JavaScript is used to access the input field and task list, create new task elements using `createElement()`, and add them to the DOM using `appendChild()`. Users can add new tasks, edit existing tasks, and delete unwanted tasks dynamically. Event listeners such as `click` are used to detect user actions and perform the required operations. The `textContent` property is used to display or modify task text, while methods like `remove()` can delete elements from the DOM. Thus, the To-Do List demonstrates practical use of DOM traversal, DOM manipulation, event handling, and dynamic webpage updates using JavaScript.

Output :



- 4. Result/Conclusion:** The To-Do List application was successfully developed using HTML, CSS, and JavaScript. DOM traversal and manipulation were performed to dynamically add, edit, and delete tasks. The application demonstrates how JavaScript can interact with HTML elements and update webpage content dynamically without reloading the page.

Case Study No.: 07

1. Case Study Title: To Demonstrate the Events Handling in JavaScript

- a) Access and manipulate form elements using JavaScript.
- b) Validate user input (e.g., email, empty fields).
- c) Use various form events (submit, change, focus).

Create the following HTML form and use required JavaScript and events in a given form.

2. Case Study Program Code:

```
<> Case study 7.html > ...
1  <html>
2  <!DOCTYPE html>
3  <html>
4  <head>
5    <title>JavaScript Events Handling</title>
6
7    <style>
8      body {
9        font-family: Arial;
10       background: #eef3f8;
11       display: flex;
12       justify-content: center;
13       padding: 40px;
14     }
15
16     .form-box {
17       width: 400px;
18       background: white;
19       padding: 25px;
20       border-radius: 10px;
21       box-shadow: 0 4px 15px #bbb;
22     }
23
24     h2 {
25       text-align: center;
26       color: #245a8d;
27     }
28
29     label {
30       display: block;
31       margin-top: 15px;
32       font-weight: bold;
33     }
34
35     input, select, textarea {
36       width: 100%;
37       padding: 10px;
38       margin-top: 5px;
39       box-sizing: border-box;
40       border: 1px solid #aaa;
41       border-radius: 5px;
42     }
43
44     textarea {
45       resize: none;
46       height: 70px;
47     }
48
49     button {
50       width: 100%;
```

```

50       width: 100%;
51       padding: 11px;
52       margin-top: 20px;
53       background: #245a8d;
54       color: white;
55       border: none;
56       border-radius: 5px;
57       cursor: pointer;
58     }
59
60     .error {
61       color: red;
62       font-size: 13px;
63     }
64
65     .success {
66       color: green;
67       margin-top: 15px;
68       font-weight: bold;
69     }
70   </style>
71 </head>
72
73 <body>
74
75   <div class="form-box">
76
77     <h2>Student Registration Form</h2>
78
79     <form id="myForm">
80
81       <label>Name</label>
82       <input type="text" id="name" placeholder="Enter your name">
83       <div id="nameError" class="error"></div>
84
85       <label>Email</label>
86       <input type="text" id="email" placeholder="Enter your email">
87       <div id="emailError" class="error"></div>
88
89       <label>Age</label>
90       <input type="number" id="age" placeholder="Enter your age">
91       <div id="ageError" class="error"></div>
92
93       <label>Course</label>
94       <select id="course">
95         <option value="">Select Course</option>
```

```

96         <option value="CSE">CSE</option>
97         <option value="IT">IT</option>
98         <option value="ECE">ECE</option>
99     </select>
100
101     <label>Address</label>
102     <textarea id="address" placeholder="Enter your address"></textarea>
103
104     <button type="submit">Submit</button>
105
106     <div id="result" class="success"></div>
107
108 </form>
109
110 </div>
111
112 <script>
113     // Accessing form elements
114     let name = document.getElementById("name");
115     let email = document.getElementById("email");
116     let age = document.getElementById("age");
117     let course = document.getElementById("course");
118     let address = document.getElementById("address");
119     let form = document.getElementById("myForm");
120
121     // Focus Event
122     name.addEventListener("focus", function() {
123         name.style.backgroundColor = "#eef7ff";
124     });
125
126     // Input Event - Name validation
127     name.addEventListener("input", function() {
128         if (name.value.trim() == "") {
129             document.getElementById("nameError").innerText =
130                 "Invalid: Name is required";
131         } else {
132             document.getElementById("nameError").innerText = "";
133         }
134     });
135
136     // Input Event - Email validation
137     email.addEventListener("input", function() {
138         let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
139         if (!pattern.test(email.value)) {
140             document.getElementById("emailError").innerText =
141                 "Invalid: Enter a valid email";
142         } else {
143             document.getElementById("emailError").innerText = "";
144         }
145     });
146
147     // Input Event - Age validation
148     age.addEventListener("input", function() {
149         if (age.value == "" || age.value < 18 || age.value > 60) {
150             document.getElementById("ageError").innerText =
151                 "Invalid: Age must be between 18 and 60";
152         } else {
153             document.getElementById("ageError").innerText = "";
154         }
155     });
156
157     course.addEventListener("change", function() {
158         document.getElementById("result").innerText =
159             "Selected Course: " + course.value;
160     });
161
162     form.addEventListener("submit", function(event) {
163         event.preventDefault();
164         let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
165         if (
166             name.value.trim() == "" ||
167             !emailPattern.test(email.value) ||
168             age.value < 18 ||
169             age.value > 60 ||
170             course.value == "" ||
171             address.value.trim() == ""
172         ) {
173             alert("Please enter valid details.");
174             return;
175         }
176
177         document.getElementById("result").innerText =
178             "Form submitted successfully! Welcome " + name.value + ".";
179     });
180
181 </script>
182
183 </body>
184 </html>
185
186

```

```

140 // Input Event - Email validation
141 email.addEventListener("input", function() {
142
143     let pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
144
145     if (!pattern.test(email.value)) {
146         document.getElementById("emailError").innerText =
147             "Invalid: Enter a valid email";
148     } else {
149         document.getElementById("emailError").innerText = "";
150     }
151
152 });
153
154 // Input Event - Age validation
155 age.addEventListener("input", function() {
156
157     if (age.value == "" || age.value < 18 || age.value > 60) {
158         document.getElementById("ageError").innerText =
159             "Invalid: Age must be between 18 and 60";
160     } else {
161         document.getElementById("ageError").innerText = "";
162     }
163
164 });
165
166 course.addEventListener("change", function() {
167
168     document.getElementById("result").innerText =
169         "Selected Course: " + course.value;
170
171 });
172
173 form.addEventListener("submit", function(event) {
174
175     event.preventDefault();
176     let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
177     if (
178         name.value.trim() == "" ||
179         !emailPattern.test(email.value) ||
180         age.value < 18 ||
181         age.value > 60 ||
182         course.value == "" ||
183         address.value.trim() == ""
184     ) {
185         alert("Please enter valid details.");
186         return;
187     }
188
189     document.getElementById("result").innerText =
190         "Form submitted successfully! Welcome " + name.value + ".";
191
192 });
193
194 </script>
195
196 </body>
197 </html>
198
199

```

```

187     document.getElementById("result").innerText =
188         "Form submitted successfully! Welcome " + name.value + ".";
189
190 });
191
192 </script>
193
194 </body>
195 </html>
196
197

```

3. Output:

~/Users/vithi/OneDrive/Desktop/flexi.py/vithikasaoji%20javascript/Case%20study%207.html

Student Registration Form

Name

Riya Sharma

Email

Riya.sharmagmail.com

Invalid: Enter a valid email

Age

10

Invalid: Age must be between 18 and 60

Course

IT

Address

25 MG Road, Nagpur

Submit

Student Registration Form

Name

Riya Sharma

Email

Riya.sharma@gmail.com

Age

20

Course

IT

Address

25 MG Road, Nagpur

Submit

Form submitted successfully! Welcome Riya Sharma .

4. Result / Conclusion:

The HTML form was successfully created using JavaScript event handling. The form elements were accessed and validated using JavaScript, and focus, change, and submit events were successfully implemented.